

# Bulgarian solitaire, Cell 3: A General Upper Bound

Team submission (task B-C3-017; accepted artefact B-C3-007)

## Statement (Cell 3, verbatim)

**Definitions (problem statement).** A *partition* of a positive integer  $n$  is a weakly decreasing sequence  $\lambda = (\lambda_1, \dots, \lambda_s)$  of positive integers with  $\lambda_1 + \dots + \lambda_s = n$  ( $s$  piles). The *shift*  $B(\lambda)$  is the partition whose parts are the positive numbers among  $\lambda_1 - 1, \dots, \lambda_s - 1$  together with one extra part equal to  $s$ .  $\lambda$  is *cyclic* if  $B^i(\lambda) = \lambda$  for some  $i \geq 1$ ;  $d_B(\lambda) = \min\{i \geq 0 : B^i(\lambda) \text{ is cyclic}\}$ ,  $D_B(n) = \max\{d_B(\lambda) : \lambda \text{ a partition of } n\}$ .  $T_k = k(k+1)/2$ ,  $\delta_k = (k, k-1, \dots, 2, 1)$ ; the *rank* of  $n$  is the unique  $k$  with  $T_{k-1} < n \leq T_k$ .

**Cell 3 (verbatim).** Now the numbers strictly between two consecutive triangular numbers.

- (a) Prove that for every  $k \geq 4$  and every non-triangular  $n$  with  $T_{k-1} < n < T_k$ ,

$$D_B(n) \leq k^2 - 2k - 1.$$

- (b) Determine  $D_B(T_k - 1)$  exactly.  
(c) Determine also, for that  $n$ , which partitions attain the maximum.

## Status

**CELL STATUS: SOLVED. CLAIM STATUS: PROVED.**

## Answers

**Answer.** Write  $\nu_k := (k+1, k-1, k-2, \dots, 3, 1)$  and  $\lambda_k^* := (k-1, k-2, k-2, k-3, \dots, 2, 1, 1)$ .

- (a) For every  $k \geq 4$ , every  $n$  with  $T_{k-1} < n < T_k$  and every partition  $\lambda \vdash n$ :  $d_B(\lambda) \leq k^2 - 2k - 1$  (Theorem 5.1; the case  $k = 4$  is done by hand).  
(b)  $D_B(T_k - 1) = k^2 - 2k - 1$  for every  $k \geq 4$ , attained by  $\lambda_k^*$  (7.1, 6.4). Small  $k$  separately:  $k = 1$ :  $T_1 - 1 = 0$  is not a positive integer, nothing to determine (7.2);  $k = 2$ :  $D_B(2) = 0$  (7.3);  $k = 3$ :  $D_B(5) = 3$  (7.4), which differs from  $3^2 - 2 \cdot 3 - 1 = 2$ .  
(c) (Proved criterion.) For every  $k \geq 4$  the partitions  $\lambda \vdash T_k - 1$  with  $d_B(\lambda) = D_B(T_k - 1)$  are exactly those with

$$B^{k^2-2k-2}(\lambda) = \nu_k$$

(Theorem 9.3). This characterises the maximiser set exactly; it is not a closed-form list (see Limitations). For  $k = 4$  the set is  $\{(3, 2, 2, 1, 1)\}$  and for  $k = 5$  it has 6 elements, listed in 9.4 (computed). For  $k = 2$  the maximisers are (2) and (1, 1) (7.3); for  $k = 3$  only (1, 1, 1, 1, 1) (7.4).

## Cited results versus our contribution

- **Method (cited).** The method follows J. R. Griggs and C.-C. Ho, *The cycling of partitions and compositions under repeated shifts*, Adv. Appl. Math. **21** (1998) 205–227, DOI 10.1006/aama.1998.0597 (preprint of 2 March 1998 at <https://people.math.sc.edu/griggs/cycling.pdf>) Sections 3–4: Prop. 3.2, Lemmas 3.3–3.6, Lemma 4.3 and **Theorem 4.4** ((1)  $D_B(n) \leq k^2 - 2k - 1$  for non-triangular  $n$ ,  $k \geq 4$ ; (2) equality for  $r = k - 1$  with the witness  $(k - 1, k - 2, k - 2, k - 3, \dots, 2, 1, 1)$ ). According to the source table of the accepted artefact, Griggs–Ho handle  $k = 4$  of Thm 4.4(1) by a computer table (their Fig. 1), and give Thm 4.4(2) only as “imitating the proof of Theorem 3.1”. **Every lemma used is re-proved in full below**; nothing is used from Griggs–Ho without proof.
- **Earlier cell (used as an assumption).** The gated Cell 1 classification of cyclic partitions: for  $k \geq 2$ ,  $n = T_{k-1} + r$ ,  $1 \leq r \leq k$ , a partition  $\mu \vdash n$  is cyclic iff  $\mu = \mu(e)$  for some  $e \in \{0, 1\}^k$  with  $\sum e_i = r$  (notation in Step 0). This is the earlier cell’s result (per Griggs–Ho Thm 2.1 it goes back to J. Brandt, *Cycles of partitions*, Proc. Amer. Math. Soc. **85** (1982) 483–486). It is referred to below as “the Cell 1 result”.
- **Our contribution.** The “pile/row” write-up of every lemma; the steps that Griggs–Ho leave as “similar”, “continue this process”, “imitating the proof of Theorem 3.1” written out; the case  $k = 4$  of (a) by hand; the written proof of the witness (Step 6); part (c) (Step 9); the small cases  $k = 1, 2, 3$  of (b). The characterisation of the maximiser set in (c) was not found in the sources searched by the accepted artefact (Griggs–Ho 1998; the Meštrović 2026 survey, arXiv:2607.17194, via a fetch summary; the Drensky 2015 and Harris–Nguyen 2023 abstracts; OEIS search).

## Proof

Throughout,  $\lambda^{(i)} := B^i(\lambda)$ .

### Step 0. Consequences of the Cell 1 result

Let  $k \geq 2$ ,  $n = T_{k-1} + r$ ,  $1 \leq r \leq k$ . For  $e \in \{0, 1\}^k$  with  $\sum e_i = r$  put  $\mu(e) = (k - 1 + e_1, k - 2 + e_2, \dots, 1 + e_{k-1}, e_k)$  (final 0 dropped). By the Cell 1 result,  $\mu \vdash n$  is cyclic iff  $\mu = \mu(e)$  for such  $e$ .

**0.1 (F1).** A cyclic  $\mu(e)$  has  $k - 1 + e_k \in \{k - 1, k\}$  parts (entries  $k - i + e_i \geq 1$  for  $i \leq k - 1$ , last entry  $e_k$ ), its  $i$ -th part is  $k - i + e_i \leq k + 1 - i$  for  $i \leq k$ , and it has no  $(k + 1)$ -st part. In particular all parts are  $\leq k$ .

**0.2 (F2).**  $B(\mu(e)) = \mu(e')$  with  $e' = (e_k, e_1, \dots, e_{k-1})$ .

*Proof.*  $\mu(e)$  has  $s = k - 1 + e_k$  parts. Subtracting 1 from the parts  $k - i + e_i$  ( $1 \leq i \leq k - 1$ ) gives  $k - (i + 1) + e_i$ , which is 0 only for  $i = k - 1, e_{k-1} = 0$ ; subtracting 1 from the part  $e_k$  (if present) gives 0. Adding the part  $s = k - 1 + e_k$ : the parts of  $B(\mu(e))$  are the positive entries of  $(k - 1 + e_k, k - 2 + e_1, k - 3 + e_2, \dots, 0 + e_{k-1})$ , which is the sequence of  $\mu(e')$ ; since  $e'$  has sum  $r$ , this is again of the cyclic form.  $\square$

**0.3 (F3).** If  $\mu$  is cyclic then  $B^j(\mu)$  is cyclic for all  $j \geq 0$  (by 0.2 and induction). Hence for any  $\lambda$ :  $\lambda^{(i)} := B^i(\lambda)$  is cyclic for all  $i \geq d_B(\lambda)$ , and  $d_B(\lambda) \leq t$  whenever  $\lambda^{(t)}$  is cyclic.

**0.4 (F4, recognition).** Let  $\mu \vdash n$  have all parts  $\leq k$  and let  $\mu'_h = \#\{i : \mu_i \geq h\}$ . If  $\mu'_h \in \{k-h, k-h+1\}$  for  $1 \leq h \leq k$ , then  $\mu$  is cyclic.

*Proof.*  $\mu'_1 \leq k$ , so  $\mu$  has at most  $k$  parts. For  $1 \leq i \leq k-1$  put  $h = k-i \in [1, k-1]$ :  $\mu'_h \geq k-h = i$ , so  $\mu_i \geq k-i$ . For  $2 \leq i \leq k$  put  $h = k-i+2 \in [2, k]$ :  $\mu'_h \leq k-h+1 = i-1 < i$ , so  $\mu_i < h$ , i.e.  $\mu_i \leq k-i+1$ ; for  $i=1$ ,  $\mu_1 \leq k$  by hypothesis. So  $e_i := \mu_i - (k-i) \in \{0, 1\}$  for  $1 \leq i \leq k$  (for  $i=k$ :  $0 \leq \mu_k \leq 1$ ),  $\sum_i e_i = n - T_{k-1} = r$ , and  $\mu = \mu(e)$ .  $\square$

## Step 1. The pile/row model (bookkeeping diagram of Griggs–Ho, Sec. 3)

Fix  $\lambda = (\lambda_1, \dots, \lambda_s) \vdash n$ . Put  $\lambda^{(i)} = B^i(\lambda)$  and  $c_{i+1} =$  number of parts of  $\lambda^{(i)}$  ( $i \geq 0$ ); so  $c_1 = s$  and  $c_i \geq 1$ .

*Piles.* Original pile  $m$  ( $1 \leq m \leq s$ ) is *alive* at rows  $1, \dots, \lambda_m$ . Born pile  $j$  ( $j \geq 1$ ) is alive at rows  $j+1, \dots, j+c_j$ . The *end row* of a pile is its last alive row ( $\lambda_m$ , resp.  $j+c_j$ ).  $R_r =$  set of piles alive at row  $r$ ;  $z_r =$  number of piles with end row  $r$ .

**1.1 Lemma.** For every  $i \geq 0$  the parts of  $\lambda^{(i)}$  are in bijection with  $R_{i+1}$ , the pile  $P$  giving the part  $\text{end}(P) - i$ . In particular  $|R_{i+1}| = c_{i+1}$ .

*Proof.* Induction on  $i$ .  $i=0$ :  $R_1$  consists of the original piles (a born pile  $j \geq 1$  is alive only from row  $j+1 \geq 2$ ), and pile  $m$  gives  $\lambda_m - 0$ . Step  $i \rightarrow i+1$ : the parts of  $B(\lambda^{(i)})$  are  $a-1$  for the parts  $a \geq 2$  of  $\lambda^{(i)}$ , plus one part  $c_{i+1}$ . A pile  $P \in R_{i+1}$  lies in  $R_{i+2}$  iff  $\text{end}(P) \geq i+2$  iff its part  $a = \text{end}(P) - i \geq 2$ , and then gives  $a-1 = \text{end}(P) - (i+1)$ . The born pile  $i+1$  is alive at rows  $i+2, \dots, i+1+c_{i+1}$ , so it is in  $R_{i+2}$  and gives  $c_{i+1}$ . No other pile is in  $R_{i+2}$ : born piles  $j \geq i+2$  start at row  $j+1 \geq i+3$ , and a pile alive at row  $i+2$  but not at row  $i+1$  must start at row  $i+2$ , i.e. be born pile  $i+1$ .  $\square$

**1.2 Corollary.** (1)  $R_{r+1} = (R_r \setminus E_r) \cup \{\text{born pile } r\}$ , where  $E_r =$  piles with end row  $r$  (all in  $R_r$ ); hence  $c_{r+1} = c_r - z_r + 1$ , i.e.  $z_r = c_r + 1 - c_{r+1}$ . (2) (GH Prop. 3.2(1))  $c_{r+1} \leq c_r + 1$ . (3) A pile alive at rows  $a$  and  $b > a$  is alive at every row in  $[a, b]$ ; a pile in  $R_a$  that is not in  $E_a \cup E_{a+1} \cup \dots \cup E_{b-1}$  is in  $R_b$ .

*Proof.* (1) from 1.1 ( $R_{r+1} =$  piles of  $R_r$  with end  $\geq r+1$ , plus born pile  $r$ ); (2) since  $z_r \geq 0$ ; (3) alive rows of a pile form an interval.  $\square$

**1.3 Terminology.** A *pattern*  $(x; p, L)$  with  $x \geq 2$ ,  $p \geq 1$ ,  $L \geq 2$  means  $(c_p, c_{p+1}, \dots, c_{p+L}) = (x-1, x, \dots, x, x+1)$ , i.e.  $c_p = x-1$ ,  $c_{p+1} = \dots = c_{p+L-1} = x$ ,  $c_{p+L} = x+1$ . By 1.2(1) this is equivalent to  $c_p = x-1$  together with  $z_p = 0$ ,  $z_{p+m} = 1$  ( $1 \leq m \leq L-2$ ),  $z_{p+L-1} = 0$ .

**1.4 (Sandwich, GH Prop. 3.2(3)).** If  $i < j$  and  $c_i < x < c_j$ , there is a pattern  $(x; p, q-p)$  with  $i \leq p < q \leq j$ .

*Proof.* Let  $p = \max\{m \in [i, j-1] : c_m \leq x-1\}$  (exists:  $m=i$ ). Then  $c_m \geq x$  for  $p < m \leq j$ .  $x \leq c_{p+1} \leq c_p + 1 \leq x$  gives  $c_p = x-1$ ,  $c_{p+1} = x$ . Let  $q = \min\{m \in (p, j] : c_m \geq x+1\}$  (exists:  $m=j$ );  $q \geq p+2$  since  $c_{p+1} = x$ . For  $p < m < q$ ,  $x \leq c_m \leq x$ ; and  $x+1 \leq c_q \leq c_{q-1} + 1 = x+1$ .  $\square$

## Step 2. Descent lemma (GH Lemmas 3.5 and 3.6, re-proved and merged)

**2.1 Lemma.** Let  $(x; p, L)$  be a pattern with  $p \geq x+1$ . Then  $L \geq 3$  and there is a pattern  $(y; p', L')$  with  $2 \leq y \leq x$ ,  $p-x \leq p' \leq p-2$ ,  $2 \leq L' \leq L-1$ .

*Proof.* (a) *Choice of  $p'$ .*  $p \geq x+1 \geq 3$ . Born pile  $p-1$  is in  $R_p$  ( $c_{p-1} \geq 1$ ). The born piles  $1, \dots, p-1$  are  $p-1 \geq x$  piles while  $|R_p| = c_p = x-1$ , so some born pile  $i \leq p-1$  is not in  $R_p$ ,

and  $i \leq p - 2$ . Let  $p' = \max\{i \leq p - 2 : \text{born pile } i \notin R_p\} \geq 1$ . Born piles  $p' + 1, \dots, p - 1$  are in  $R_p$ ; they are  $p - 1 - p'$  distinct elements of  $R_p$ , so  $p - 1 - p' \leq x - 1$ , i.e.  $y := p - p' \leq x$ ; also  $y \geq 2$ .

(b) *Two exact values.* Pile  $p'$  starts at row  $p' + 1 \leq p - 1$  and is not alive at row  $p$ , so  $p' + c_{p'} \leq p - 1$ :  $c_{p'} \leq y - 1$ . Pile  $p' + 1 \in R_p$ , and  $z_p = 0$ , so it is in  $R_{p+1}$ :  $p' + 1 + c_{p'+1} \geq p + 1$ ,  $c_{p'+1} \geq y$ . With 1.2(2):  $c_{p'} = y - 1$ ,  $c_{p'+1} = y$ , and pile  $p' + 1$  has end row exactly  $p + 1$ .

(c) *Induction.* For  $1 \leq m \leq L - 1$  let  $H(m)$ :  $c_{p'+1} = \dots = c_{p'+m} = y$  and, for  $1 \leq l \leq m$ , pile  $p' + l$  has end row  $p + l$ .  $H(1)$  holds by (b).

- If  $H(L - 1)$  held, pile  $p' + L - 1$  would end at row  $p + L - 1$ , contradicting  $z_{p+L-1} = 0$ . So the induction below must stop earlier. (For  $L = 2$ :  $H(1) = H(L - 1)$  holds, contradiction; hence  $L \geq 3$ . This is GH Lemma 3.6.)
- Assume  $H(m)$  with  $m \leq L - 2$ . Let  $j = p' + m + 1 \leq p + m - 1$  (as  $p' \leq p - 2$ ). *Claim:* pile  $j \in R_{p+m+1}$ . The piles ending at rows  $p, \dots, p + m$ : none at row  $p$  ( $z_p = 0$ ), and for  $1 \leq l \leq m$  exactly one at row  $p + l$  ( $z_{p+l} = 1$  as  $p + l \leq p + L - 2$ ), which by  $H(m)$  is pile  $p' + l \neq j$ . If  $j \leq p - 1$ , pile  $j \in R_p$  by choice of  $p'$ ; if  $p \leq j \leq p + m - 1$ , pile  $j \in R_{j+1}$  with  $j + 1 \in [p + 1, p + m]$ . In both cases, by 1.2(3),  $j$  is alive at row  $p + m + 1$ . Hence  $c_j \geq p + m + 1 - j = y$ , and  $c_j \leq c_{j-1} + 1 = y + 1$ .
- If  $c_j = y + 1$ :  $(c_{p'}, \dots, c_{p'+m+1}) = (y - 1, y, \dots, y, y + 1)$  is a pattern  $(y; p', m + 1)$  with  $2 \leq m + 1 \leq L - 1$ . Done.
- If  $c_j = y$ : pile  $j$  ends at row  $j + y = p + m + 1$ , so  $H(m + 1)$  holds.

Since  $H(L - 1)$  is impossible, the first alternative occurs for some  $m \leq L - 2$ .  $\square$

**2.2 Corollary (GH counting in proof of Thm 3.7/4.4).** If  $(x; p, L)$  is a pattern with  $x \leq X$ , then  $p \leq (L - 1)X$ .

*Proof.* Build a chain  $P_0 = (x_0; p_0, L_0) = (x; p, L)$ , and while  $p_j \geq x_j + 1$  let  $P_{j+1}$  be the pattern of 2.1. Then  $x_{j+1} \leq x_j \leq X$ ,  $L_{j+1} \leq L_j - 1$ ,  $p_j \leq p_{j+1} + x_j \leq p_{j+1} + X$ . Since every pattern has  $L \geq 2$  and a pattern with  $L = 2$  has  $p \leq x$  (2.1), the chain stops at some  $P_f$  with  $p_f \leq x_f \leq X$  and  $f \leq L_0 - 2$ . So  $p \leq fX + X \leq (L - 1)X$ .  $\square$

### Step 3. A long pattern forces a long original pile (GH Lemma 3.4)

**3.1 Lemma.** Let  $k \geq 3$  and let  $(k - 1; p, k)$  be a pattern, i.e.  $(c_p, \dots, c_{p+k}) = (k - 2, k - 1, \dots, k - 1, k)$ . Then  $p + k \leq n + 1$ , with equality only if  $\lambda = (1^n)$ .

*Proof.* Here  $z_p = 0$ ,  $z_{p+m} = 1$  ( $1 \leq m \leq k - 2$ ),  $z_{p+k-1} = 0$ . For  $j \in [p + 1, p + k - 1]$ ,  $c_j = k - 1$ , so pile  $j$  is alive at rows  $j + 1, \dots, j + k - 1 \ni p + k$ . As  $|R_{p+k}| = c_{p+k} = k$ ,  $R_{p+k} = \{p + 1, \dots, p + k - 1\} \cup \{X\}$  for one further pile  $X$ ;  $X$  is not born pile  $p$  (end row  $p + c_p = p + k - 2$ ), nor born  $\geq p + k$  (not yet alive), so  $X$  is original or born  $j \leq p - 1$ . By 1.2(1) with  $z_{p+k-1} = 0$ :  $R_{p+k-1} = R_{p+k} \setminus \{p + k - 1\} = \{p + 1, \dots, p + k - 2\} \cup \{X\}$ . With  $z_{p+k-2} = 1$  ( $k \geq 3$ ) and pile  $p \in R_{p+k-2}$  ending at row  $p + k - 2$  (it is alive at rows  $p + 1, \dots, p + k - 2$ , nonempty as  $k \geq 3$ ),  $E_{p+k-2} = \{p\}$  and  $R_{p+k-2} = \{p, \dots, p + k - 3\} \cup \{X\}$ .

If  $X$  is original pile  $m$ :  $\lambda_m \geq p + k$  and  $\lambda_m \leq n$ , so  $p + k \leq n$ .

If  $X$  is born pile  $j \leq p - 1$  with  $j \geq 2$ : pile  $j - 1$  ( $\leq p - 2$ ) is not  $X$  and not in  $\{p, \dots\}$ , so  $j - 1 \notin R_{p+k-2}$ ; it is alive at row  $j \leq p + k - 2$ , hence its end row is  $\leq p + k - 3$ :  $c_{j-1} \leq p + k - j - 2$ . But  $X = j \in R_{p+k}$  gives  $c_j \geq p + k - j$ , so  $c_j \geq c_{j-1} + 2$ , contradicting 1.2(2). Hence  $j = 1$  and  $c_1 = s \geq p + k - 1$ ;  $s \leq n$  gives  $p + k \leq n + 1$ , with equality only if  $s = n$ , i.e.  $\lambda = (1^n)$ .  $\square$

**Step 4. Structure at the entry into the cycle (GH Lemma 3.3(2) and 4.3, re-proved)**

From here on  $k \geq 3$ ,  $n = T_{k-1} + r$ ,  $1 \leq r \leq k-1$ .

**4.1 Definition/Lemma.**  $\tau := \min\{i \geq 1 : \lambda^{(i-1)} \text{ cyclic, } c_i = k-1, c_{i+1} = k\}$  exists, and  $d_B(\lambda) \leq \tau - 1$ .

*Proof.* Let  $d = d_B(\lambda)$ ,  $\lambda^{(d)} = \mu(e)$ . By 0.2,  $\lambda^{(d+j)} = \mu(e^{(j)})$  with  $e^{(j)}$  the  $j$ -fold cyclic rotation of  $e$ , whose last coordinate is  $e_{k-j \bmod k}$  (indices in  $1..k$ ). So  $c_{d+j+1} = k-1 + e_{(k-j) \bmod k}$  is periodic in  $j$  with period  $k$  and, since  $1 \leq r \leq k-1$ , takes both values  $k-1$  and  $k$ . Take  $j_0 \geq d+1$  with  $c_{j_0} = k-1$  and  $j_1 = \min\{j > j_0 : c_j = k\}$ ; then  $i = j_1 - 1 \geq d+1$  has  $c_i = k-1$ ,  $c_{i+1} = k$ , and  $\lambda^{(i-1)}$  is cyclic by 0.3. So  $\tau$  exists, and  $d \leq \tau - 1$  by 0.3.  $\square$

**4.2 Lemma.** If  $\tau \geq k+1$ , at least one holds:

- (i) there is a pattern  $(k; p, L)$  with  $\tau - k \leq p$  and  $p + L \leq \tau - 1$ ;
- (ii) there is a pattern  $(k-1; p, L)$  with  $\tau - k + 1 \leq p$  and  $p + L \leq \tau + 1$ .

*Proof.*  $|R_\tau| = c_\tau = k-1$ , born pile  $\tau-1 \in R_\tau$ , and born piles  $\tau-k, \dots, \tau-1$  (all  $\geq 1$ ) are  $k$  piles, so one of them with index  $\leq \tau-2$  is not in  $R_\tau$ . Let  $i = \max\{j \in [\tau-k, \tau-2] : \text{born pile } j \notin R_\tau\}$ . As in 2.1(b) (with  $z_\tau = c_\tau + 1 - c_{\tau+1} = 0$ ):  $c_i \leq \tau-1-i$ , pile  $i+1 \in R_{\tau+1}$  so  $c_{i+1} \geq \tau-i$ , hence  $c_i = \tau-i-1$ ,  $c_{i+1} = \tau-i$ .

*Case A:*  $i \geq \tau-k+1$ . Then  $c_i \leq k-2 < k-1 < k = c_{\tau+1}$ ; 1.4 on  $(i, \tau+1)$  gives (ii) with  $p \geq i$ .

*Case B:*  $i = \tau-k$ . Then  $c_{\tau-k} = k-1$ ,  $c_{\tau-k+1} = k$ , and born piles  $\tau-k+1, \dots, \tau-1$  (that is  $k-1 = |R_\tau|$  piles) are in  $R_\tau$ , so  $R_\tau = \{\tau-k+1, \dots, \tau-1\}$ .

B1: some  $j \in [\tau-k+2, \tau-1]$  has  $c_j \leq k-2$ : 1.4 on  $(j, \tau+1)$  with  $x = k-1$  gives (ii).

B2: some  $j \in [\tau-k+2, \tau-1]$  has  $c_j \geq k+1$ : 1.4 on  $(\tau-k, j)$  with  $x = k$  gives (i) (with  $p \geq \tau-k$ ,  $p+L \leq j \leq \tau-1$ ).

B3: otherwise  $c_j \in \{k-1, k\}$  for all  $j \in [\tau-k, \tau-1]$ . We show  $\lambda^{(\tau-k-1)}$  is cyclic; together with  $c_{\tau-k} = k-1$ ,  $c_{\tau-k+1} = k$  and  $\tau-k \geq 1$  this contradicts the minimality of  $\tau$ . By 1.1, the parts of  $\lambda^{(\tau-k-1)}$  correspond to  $R_{\tau-k}$ , pile  $P$  giving  $\text{end}(P) - \tau + k + 1$ . No pile of  $R_{\tau-k}$  is in  $R_\tau = \{\tau-k+1, \dots, \tau-1\}$ , since born pile  $j$  is alive only from row  $j+1 > \tau-k$ ; so by 1.2(3) every  $P \in R_{\tau-k}$  has end row in  $[\tau-k, \tau-1]$ , and its part  $h := \text{end}(P) - \tau + k + 1 \in [1, k]$ . Conversely a pile with end row  $\rho \in [\tau-k, \tau-1]$  is in  $R_{\tau-k}$  unless it starts after row  $\tau-k$ , i.e. is a born pile  $j \geq \tau-k$ ; among these, pile  $\tau-k$  ends at  $\tau-k + c_{\tau-k} = \tau-1$ , piles  $\tau-k+1, \dots, \tau-1$  are in  $R_\tau$  (end  $\geq \tau$ ), piles  $j \geq \tau$  end at  $\geq \tau+1$ . So the number of parts equal to  $h$  is  $a_h = z_{\tau-k-1+h} - [h = k]$ , and for  $1 \leq h \leq k$ , using  $z_\rho = c_\rho + 1 - c_{\rho+1}$  and telescoping,

$$\mu'_h := \sum_{g=h}^k a_g = \sum_{g=h}^k (c_{\tau-k-1+g} + 1 - c_{\tau-k+g}) - 1 = c_{\tau-k-1+h} - c_\tau + (k-h) = c_{\tau-k-1+h} + 1 - h,$$

using  $c_\tau = k-1$ . As  $\tau-k-1+h \in [\tau-k, \tau-1]$ ,  $c_{\tau-k-1+h} \in \{k-1, k\}$ , so  $\mu'_h \in \{k-h, k-h+1\}$ ; all parts are  $\leq k$ , and  $\lambda^{(\tau-k-1)} \vdash n$ . By 0.4 it is cyclic.  $\square$

**Step 5. Theorem (target (a)); GH Thm 4.4(1), with  $k = 4$  done by hand**

**5.1 Theorem.** Let  $k \geq 4$ ,  $T_{k-1} < n < T_k$ ,  $\lambda \vdash n$ . Then  $\tau - 1 \leq k^2 - 2k - 1$ , hence  $d_B(\lambda) \leq k^2 - 2k - 1$ . Moreover, if  $\tau - 1 = k^2 - 2k - 1$  then  $\tau \geq k + 1$  and: for  $k \geq 5$  case (i) of 4.2 holds; for  $k = 4$  case (i) holds or case (ii) holds with  $L = k$  and  $p = \tau - k + 1$ .

*Proof.*  $n = T_{k-1} + r$ ,  $1 \leq r \leq k - 1$ . If  $\tau \leq k$ :  $\tau - 1 \leq k - 1 < k^2 - 2k - 1$  ( $k \geq 4$ ). Else 4.2 applies.

(i) Pattern  $(k; p, L)$ ,  $\tau - k \leq p$ ,  $p + L \leq \tau - 1$ , so  $L \leq k - 1$ . If  $L = k - 1$ :  $p = \tau - k$  and  $c_{\tau-1} = k + 1$ ; but born pile  $\tau - 1$  gives the part  $c_{\tau-1}$  of  $\lambda^{(\tau-1)}$  (1.1), which is cyclic, so  $c_{\tau-1} \leq k$  by 0.1: contradiction. So  $L \leq k - 2$ , and 2.2 with  $X = k$  gives  $p \leq (k - 3)k$ . Then  $\tau - 1 \leq p + k - 1 \leq k^2 - 2k - 1$ .

(ii) Pattern  $(k - 1; p, L)$ ,  $\tau - k + 1 \leq p$ ,  $p + L \leq \tau + 1$ , so  $L \leq k$ .

- $L = k$ :  $p = \tau - k + 1$ ; by 3.1,  $\tau - 1 = p + k - 2 \leq n - 1 \leq T_k - 2$ . For  $k \geq 5$ :  $T_k - 2 < k^2 - 2k - 1 \iff k^2 - 5k + 2 > 0$ , true. For  $k = 4$ :  $n \leq 9$ ,  $\tau - 1 \leq n - 1 \leq 8$ , and  $= 8$  forces  $n = 9$  and equality in 3.1, i.e.  $\lambda = (1^9)$ ; otherwise  $\tau - 1 \leq 7 = k^2 - 2k - 1$ . For  $\lambda = (1^9)$  directly:

$$(1^9) \rightarrow (9) \rightarrow (8, 1) \rightarrow (7, 2) \rightarrow (6, 2, 1) \rightarrow (5, 3, 1) \rightarrow (4, 3, 2) \rightarrow (3, 3, 2, 1)$$

(each arrow is one application of  $B$ , checked from the definition), and  $(4, 3, 2) = \mu(1, 1, 1, 0)$  is cyclic ( $k = 4, r = 3$ , Cell 1 result). So  $\lambda^{(6)}$  is cyclic,  $c_7 = 3$  (parts of  $(4, 3, 2)$ ),  $c_8 = 4$  (parts of  $(3, 3, 2, 1)$ ), hence  $\tau \leq 7$  and  $\tau - 1 \leq 6 < 7$ ; so  $\lambda = (1^9)$  does not reach this case with  $\tau - 1 = 8$  at all, and in every sub-case  $\tau - 1 \leq 7$ .

- $L = k - 1$ :  $p \in \{\tau - k + 1, \tau - k + 2\}$ . If  $p = \tau - k + 1$  then  $c_\tau = c_{p+L} = k \neq k - 1$ : impossible. If  $p = \tau - k + 2$  then  $c_p = k - 2$  and born pile  $p$  ends at row  $p + k - 2 = \tau$ , so  $z_\tau \geq 1$ , but  $z_\tau = c_\tau + 1 - c_{\tau+1} = 0$ : impossible.
- $L \leq k - 2$ : 2.2 with  $X = k - 1$ :  $p \leq (L - 1)(k - 1) \leq (k - 3)(k - 1)$ . From  $\tau - k + 1 \leq p$ :  $\tau - 1 \leq p + k - 2 \leq (k - 3)(k - 1) + k - 2 = k^2 - 3k + 1 < k^2 - 2k - 1$  (as  $k > 2$ ).

In all cases  $\tau - 1 \leq k^2 - 2k - 1$ , and  $d_B(\lambda) \leq \tau - 1$  by 4.1. The “moreover” is read off:  $\tau \leq k$  gives  $\tau - 1 < k^2 - 2k - 1$ ; in case (ii),  $L = k - 1$  is impossible,  $L \leq k - 2$  gives strict inequality, and  $L = k$  gives strict inequality for  $k \geq 5$  (it forces  $p = \tau - k + 1$ ).  $\square$

**Step 6. The witness (target (b), lower bound); GH Thm 4.4(2) “imitating Thm 3.1”, written out**

Diagonal picture: cell  $(i, j)$  of a diagram lies on diagonal  $i + j - 1$  (rows = parts). Let  $k \geq 3$ .

**6.1 Configurations.** For  $H \subseteq \{1, \dots, k\}$  and  $\pi \in \{1, \dots, k + 1\}$  let  $\ell_i(H, \pi) = (k - i)^+ + [i \leq k, i \notin H] + [\pi = i]$  for  $1 \leq i \leq k + 1$ . Condition (S): if  $\pi \leq k$  then  $\pi \notin H$ ; if  $\pi \geq 2$  then  $\pi - 1 \notin H$ .

Under (S),  $\ell_1 \geq \ell_2 \geq \dots \geq \ell_{k+1} \geq 0$ : for  $i \leq k - 1$ ,  $\ell_i - \ell_{i+1} = 1 + [i \notin H] + [\pi = i] - [i + 1 \notin H] - [\pi = i + 1]$  is negative only if  $i \in H$  and  $\pi = i + 1$ , excluded by (S);  $\ell_k - \ell_{k+1} = [k \notin H] + [\pi = k] - [\pi = k + 1]$  is negative only if  $k \in H, \pi = k + 1$ , excluded by (S). So the positive  $\ell_i$  form a partition  $\Lambda(H, \pi)$ ; the same holds for  $\Lambda(H) :=$  positive entries of  $\ell_i(H) = (k - i)^+ + [i \leq k, i \notin H]$  (no  $\pi$ ).

Rotation:  $H + 1 := \{h + 1 : h \in H, h < k\} \cup \{1 : k \in H\}$ ,  $\pi^+ := \pi + 1$  if  $\pi \leq k$ ,  $\pi^+ := 1$  if  $\pi = k + 1$ .

**6.2 Lemma.** Assume (S). (1) If not  $(\pi = 1 \text{ and } k \in H)$  then  $B(\Lambda(H, \pi)) = \Lambda(H + 1, \pi^+)$  and  $(H + 1, \pi^+)$  satisfies (S). (2) If  $\pi = 1$  and  $k \in H$  then  $B(\Lambda(H, \pi)) = \Lambda((H + 1) \setminus \{1\})$ .

*Proof.* Number of parts:  $\ell_i \geq k - i \geq 1$  for  $i \leq k - 1$ ;  $\ell_k = [k \notin H] + [\pi = k]$  is positive iff  $k \notin H$  (if  $\pi = k$  then  $k \notin H$  by (S));  $\ell_{k+1} = [\pi = k + 1]$ . So  $s = k - 1 + [k \notin H] + [\pi = k + 1]$ . The parts of  $B$  are  $s$  and the positive values  $\ell_i - 1$ .

Compare with  $\ell'_j := \ell_j(H + 1, \pi^+)$ :  $\ell'_1 = (k - 1) + [1 \notin H + 1] + [\pi^+ = 1] = (k - 1) + [k \notin H] + [\pi = k + 1] = s$ . For  $1 \leq i \leq k - 1$ :  $\ell'_{i+1} = (k - i - 1) + [i + 1 \notin H + 1] + [\pi^+ = i + 1] = (k - i - 1) + [i \notin H] + [\pi = i] = \ell_i - 1$ . For  $i = k$ :  $\ell'_{k+1} = [\pi^+ = k + 1] = [\pi = k]$ , while  $\ell_k - 1 = [k \notin H] + [\pi = k] - 1$ , which is  $[\pi = k]$  if  $k \notin H$ , and  $\leq 0$  (no part) with  $[\pi = k] = 0$  if  $k \in H$  (by (S)); so the positive values agree.  $\ell_{k+1} - 1 \leq 0$  gives no part. Hence the multiset of parts of  $B(\Lambda(H, \pi))$  is the multiset of positive  $\ell'_j$ .

(1) (S) for  $(H + 1, \pi^+)$ : if  $\pi^+ \leq k$ : for  $\pi \leq k - 1$ ,  $\pi + 1 \in H + 1 \iff \pi \in H$ , false by (S); for  $\pi = k + 1$ ,  $1 \in H + 1 \iff k \in H$ , false by (S) ( $\pi - 1 = k$ ). If  $\pi^+ \geq 2$  (so  $\pi \leq k$ ):  $\pi^+ - 1 = \pi \in H + 1$  iff ( $\pi \geq 2$  and  $\pi - 1 \in H$ ) or ( $\pi = 1$  and  $k \in H$ ); the first is excluded by (S), the second by hypothesis. So by 6.1  $\ell'$  is weakly decreasing and  $B(\Lambda(H, \pi)) = \Lambda(H + 1, \pi^+)$ .

(2) Here  $\pi = 1$ ,  $1 \notin H$  (S),  $k \in H$ :  $s = k - 1$ ,  $\ell_1 - 1 = k$ ,  $\ell' = (k - 1, k, \ell_2 - 1, \dots)$  as multiset. The partition is the sorted multiset  $(k, k - 1, \ell_2 - 1, \ell_3 - 1, \dots)$  (note  $\ell_2 - 1 \leq k - 2$ ). On the other hand  $\ell_j((H + 1) \setminus \{1\})$  equals  $k - 1 + 1 = k$  for  $j = 1$  ( $1 \notin (H + 1) \setminus \{1\}$ ),  $k - 2 + [2 \notin H + 1] = k - 2 + [1 \notin H] = k - 1$  for  $j = 2$ , and  $\ell'_j$  for  $3 \leq j \leq k$  (as  $\pi^+ = 2$ ), and 0 for  $j = k + 1$  ( $= \ell'_{k+1} = [\pi = k] = 0$ ). Both are the same multiset, and  $\ell((H + 1) \setminus \{1\})$  is decreasing by 6.1 (no  $\pi$ ).  $\square$

**6.3 Non-cyclicity.** Let  $|H| \geq 1$ , so  $|\Lambda(H, \pi)| = T_{k-1} + (k - |H|) + 1 \in (T_{k-1}, T_k]$  has rank  $k$  and 0.1 applies. Then  $\Lambda(H, \pi)$  (any  $\pi$ , under (S)) is not cyclic: if  $\pi \leq k$ , its  $\pi$ -th part is  $(k - \pi) + 1 + 1 > k + 1 - \pi$  ( $\pi \notin H$  by (S)), violating 0.1; if  $\pi = k + 1$  it has a  $(k + 1)$ -st part, violating 0.1.  $\Lambda(H)$  with  $|\Lambda(H)| = T_{k-1} + (k - |H|)$  equals  $\mu(e)$  with  $e_i = [i \notin H]$ , hence is cyclic (Cell 1 result,  $r = k - |H| \geq 1$  when  $H \neq \{1..k\}$ ).

**6.4 Proposition.** For  $k \geq 3$ ,  $\lambda_k^* := \Lambda(\{1, 2\}, k + 1) = (k - 1, k - 2, k - 2, k - 3, \dots, 2, 1, 1) \vdash T_k - 1$  satisfies  $d_B(\lambda_k^*) = k^2 - 2k - 1$  and  $B^{k^2 - 2k - 2}(\lambda_k^*) = \nu_k$ .

*Proof.* Rows:  $\ell_1 = k - 1$ ,  $\ell_2 = k - 2$ ,  $\ell_i = k - i + 1$  ( $3 \leq i \leq k$ ),  $\ell_{k+1} = 1$ : this is  $\lambda_k^*$ , of size  $T_{k-1} + (k - 2) + 1 = T_k - 1$ . (S):  $\pi = k + 1$  needs  $k \notin \{1, 2\}$ , true for  $k \geq 3$ .

Let  $H_t = \{1 + t, 2 + t\} \bmod k$  (representatives in  $1..k$ ) and  $\pi_t \equiv t \pmod{k + 1}$  in  $1..k + 1$  (so  $\pi_0 = k + 1$ ). As long as no exceptional step occurs, 6.2(1) and induction give  $\lambda^{(t)} = \Lambda(H_t, \pi_t)$  with (S). The exceptional condition at time  $t$  is  $\pi_t = 1$  and  $k \in H_t$ :  $\pi_t = 1 \iff t = (k + 1)v + 1$ ,  $v \geq 0$ ; then  $t \equiv v + 1 \pmod{k}$ , and  $k \in H_t \iff t \equiv -1$  or  $-2 \pmod{k} \iff v \equiv k - 2$  or  $k - 3 \pmod{k}$ . The least  $v \geq 0$  is  $v = k - 3$ , so the first exceptional time is  $t^* = (k + 1)(k - 3) + 1 = k^2 - 2k - 2$ . For  $0 \leq t \leq t^*$ ,  $\lambda^{(t)} = \Lambda(H_t, \pi_t)$  is not cyclic (6.3). By 6.2(2),  $\lambda^{(t^*+1)} = \Lambda(H')$  with  $|H'| = 1$ , cyclic (6.3). By 0.3,  $d_B(\lambda_k^*) = t^* + 1 = k^2 - 2k - 1$ .

At  $t^*$ :  $t^* \equiv -2 \pmod{k}$ , so  $H_{t^*} = \{k - 1, k\}$ ,  $\pi_{t^*} = 1$ , rows  $\ell_1 = (k - 1) + 1 + 1 = k + 1$ ,  $\ell_i = k - i + 1$  ( $2 \leq i \leq k - 2$ ),  $\ell_{k-1} = 1$ ,  $\ell_k = \ell_{k+1} = 0$ : this is  $\nu_k = (k + 1, k - 1, \dots, 3, 1)$ .  $\square$

## Step 7. Target (b)

**7.1**  $k \geq 4$ :  $D_B(T_k - 1) = k^2 - 2k - 1$ : “ $\leq$ ” is 5.1 with  $n = T_k - 1$  (non-triangular, rank  $k$ ); “ $\geq$ ” is 6.4.

**7.2**  $k = 1$ :  $T_1 - 1 = 0$  is not a positive integer; nothing to determine.

**7.3**  $k = 2$ ,  $n = 2$ : by the Cell 1 result ( $k = 2, r = 1$ ) the cyclic partitions are  $(1 + e_1, e_2)$ ,  $e_1 + e_2 = 1$ : (2) and (1, 1), i.e. all partitions of 2.  $D_B(2) = 0$ , attained by both.

**7.4**  $k = 3, n = 5$  ( $r = 2$ ): cyclic are  $(2 + e_1, 1 + e_2, e_3)$  with two ones:  $(3, 2), (3, 1, 1), (2, 2, 1)$ . Directly:  $B(4, 1) = (3, 2)$ ;  $B(5) = (4, 1)$ ;  $B(2, 1, 1, 1) = (4, 1)$ ;  $B(1, 1, 1, 1, 1) = (5)$ . So  $d_B = 0$  on the three cyclic ones,  $d_B(4, 1) = 1$ ,  $d_B(5) = d_B(2, 1, 1, 1) = 2$ ,  $d_B(1^5) = 3$ . These are all 7 partitions of 5:  $D_B(5) = 3$  ( $\neq 3^2 - 6 - 1 = 2$ ), attained only by  $(1^5)$ .

## Step 8. (not needed) remark

For  $k = 3$  Theorem 5.1 fails ( $D_B(5) = 3 > 2$ ); GH note the same. The proof of 5.1 uses  $k \geq 4$  only in the final numeric comparisons.

## Step 9. Target (c)

**9.1 Proposition.** Let  $k \geq 4, m = k^2 - 2k - 2$ . If  $\lambda \vdash T_k - 1$  and  $B^m(\lambda) = \nu_k$ , then  $d_B(\lambda) = k^2 - 2k - 1$ .

*Proof.*  $\nu_k$  has first part  $k + 1 > k$ , so it is not cyclic (0.1). If  $\lambda^{(i)}$  were cyclic for some  $i \leq m$ , then  $\lambda^{(m)} = \nu_k$  would be cyclic (0.3). So  $d_B(\lambda) \geq m + 1$ .  $B(\nu_k)$ :  $\nu_k$  has  $1 + (k - 3) + 1 = k - 1$  parts; subtracting 1 gives  $k, k - 2, \dots, 2$  and 0 (dropped); adding  $k - 1$ :  $B(\nu_k) = (k, k - 1, \dots, 2) = \mu(1, \dots, 1, 0)$ , cyclic. So  $d_B(\lambda) = m + 1$ .  $\square$

**9.2 Proposition.** Let  $k \geq 4, \lambda \vdash T_k - 1$  with  $d_B(\lambda) = k^2 - 2k - 1$ . Then  $B^{k^2 - 2k - 2}(\lambda) = \nu_k$ .

*Proof.* By 4.1 and 5.1,  $k^2 - 2k - 1 = d_B(\lambda) \leq \tau - 1 \leq k^2 - 2k - 1$ , so  $\tau = k^2 - 2k \geq k + 1$ . *Case (i) of 4.2 holds.* For  $k \geq 5$  this is the “moreover” of 5.1. For  $k = 4$  ( $n = 9, \tau = 8$ ), by the “moreover” the only alternative is case (ii) with  $L = 4, p = \tau - k + 1 = 5$ , i.e. a pattern  $(3; 5, 4)$ ; then the proof of 3.1 (with  $p + k = 9$ ) shows that the extra pile  $X \in R_9$  is either an original pile with  $\lambda_m \geq 9$ , forcing  $\lambda = (9)$ , or born pile 1 with  $s = c_1 \geq 8$ , forcing  $\lambda \in \{(2, 1^7), (1^9)\}$ . From the chain in 5.1,  $(9) \rightarrow (8, 1) \rightarrow (7, 2) \rightarrow (6, 2, 1) \rightarrow (5, 3, 1) \rightarrow (4, 3, 2)$  gives  $d_B((9)) \leq 5$ ;  $B((2, 1^7)) = (8, 1)$  gives  $d_B((2, 1^7)) \leq 5$ ; and  $d_B((1^9)) \leq 6$ . All are  $< 7 = d_B(\lambda)$ , a contradiction; so case (i) holds for  $k = 4$  as well. So there is a pattern  $(k; p, L)$  with  $\tau - k \leq p, p + L \leq \tau - 1$ ; the proof of 5.1 showed  $L \leq k - 2$  and  $p \leq (L - 1)k$  (2.2). From  $\tau - 1 \leq p + k - 1$  and  $p \leq (L - 1)k \leq (k - 3)k = \tau - k$ :  $p = \tau - k$  and  $L = k - 2$ . So  $c_{\tau - k} = k - 1, c_{\tau - k + 1} = \dots = c_{\tau - 3} = k, c_{\tau - 2} = k + 1$ .

Born piles  $j = \tau - k + 1, \dots, \tau - 1$  are in  $R_\tau$ : for  $j \leq \tau - 3, j + c_j = j + k \geq \tau + 1$ ; for  $j = \tau - 2, j + c_j = \tau + k - 1 \geq \tau$ ; for  $j = \tau - 1$ , alive at row  $\tau$ . These are  $k - 1 = c_\tau$  piles, so they are all of  $R_\tau$ . By 1.1 the parts of  $\lambda^{(\tau - 1)}$  are  $j + c_j - (\tau - 1)$ :  $c_{\tau - 1}$  (for  $j = \tau - 1$ ),  $k$  (for  $j = \tau - 2$ ), and  $k - m'$  for  $j = \tau - 1 - m', 2 \leq m' \leq k - 2$ . Their sum is  $T_k - 1$ , so  $c_{\tau - 1} = T_k - 1 - k - (T_{k - 2} - 1) = k - 1$  and  $\lambda^{(\tau - 1)} = (k, k - 1, k - 2, \dots, 2)$ .

Now  $\lambda^{(\tau - 2)}$  has  $c_{\tau - 1} = k - 1$  parts, and its parts  $\geq 2$ , decreased by 1, together with the part  $k - 1$  form  $\lambda^{(\tau - 1)}$ . Removing one part  $k - 1$  from  $(k, k - 1, \dots, 2)$  leaves  $k, k - 2, \dots, 2$ , so the parts  $\geq 2$  of  $\lambda^{(\tau - 2)}$  are  $k + 1, k - 1, \dots, 3$  ( $k - 2$  parts), and the remaining one part is 1. So  $\lambda^{(\tau - 2)} = \nu_k, \tau - 2 = k^2 - 2k - 2$ .  $\square$

**9.3 Theorem (c),  $k \geq 4$ .** The maximisers of  $d_B$  on partitions of  $T_k - 1$  are exactly

$$\mathcal{E}_k = \{\lambda \vdash T_k - 1 : B^{k^2 - 2k - 2}(\lambda) = \nu_k\}, \quad \nu_k = (k + 1, k - 1, k - 2, \dots, 3, 1);$$

$\lambda_k^* \in \mathcal{E}_k$  (6.4).

*Proof.* 7.1, 9.1, 9.2.  $\square$

**9.4 Small cases of  $\mathcal{E}_k$**  (computed; illustration and cross-check only, the proof of 9.3 does not use it).

$k = 4$ :  $\mathcal{E}_4 = \{(3, 2, 2, 1, 1)\} = \{\lambda_4^*\}$ .  $k = 5$ :  $\mathcal{E}_5 = \{(4, 3, 3, 2, 1, 1), (4, 3, 2, 2, 2, 1), (4, 3, 2, 2, 1, 1, 1), (3, 3, 3, 2, 2, 1), (3, 3, 2, 2, 2, 1)\}$ , the first being  $\lambda_5^*$ . Both lists are printed by `code/check.py` (output `code/run.txt`) and agree with



an exhaustive exact computation of  $d_B$  on all partitions of 9 and 14 (that computation identifies cyclic partitions via the Cell 1 list).

**9.5**  $k = 2$ : maximisers  $\{(2), (1, 1)\}$  (7.3);  $k = 3$ :  $\{(1^5)\}$  (7.4).

**9.6 On “explicit functions of  $k$ ”.**  $\mathcal{E}_k$  is described dynamically (the set of partitions at depth exactly  $k^2 - 2k - 2$  above  $\nu_k$  in the  $B$ -tree).  $|\mathcal{E}_k| = 1, 6, 34, 175, 831, 3911$  for  $k = 4, \dots, 9$  (computation, not a proof of anything). The answer to (c) is therefore a proved criterion that characterises the maximiser set exactly; a closed-form list of  $\mathcal{E}_k$  is not obtained. Griggs–Ho (Thm 3.8 and remark after it) report that for the analogous triangular case even three natural necessary array conditions are not sufficient ( $k = 8$ ).

## Step 10. What is established

- PROVED (all  $k$  stated): Step 0 (from the Cell 1 result), 1.1–1.4, 2.1, 2.2, 3.1, 4.1, 4.2, Theorem 5.1 ((a) for every  $k \geq 4$ , by hand, including  $k = 4$ ), 6.4, (b) for all  $k \geq 1$  (7.1–7.4), (c) for every  $k \geq 4$  (9.3, by hand, including  $k = 4$ ) and  $k = 2, 3$  (7.3, 7.4).
- Not established: a closed-form list of  $\mathcal{E}_k$  (9.6); (c) is answered by the explicit criterion  $B^{k^2-2k-2}(\lambda) = \nu_k$ .
- Sanity only (not used): `check.py` confirms (a), (b), (c) and  $\tau - 1 \leq k^2 - 2k - 1$  for  $4 \leq k \leq 9$ , matching GH Fig. 1.

## How to verify (sanity check, not load-bearing)

`code/check.py` is stdlib-only exact integer computation. From the directory `submission/code/`:

```
/usr/bin/time -p python3 check.py 9
```

Expected output: the 34 lines recorded in `code/run.txt`, ending with ALL OK. Measured on our re-run (Python 3.14.0): real 38.62s. The proof above does not depend on this computation.

## Limitations

- (c) is answered by a proved criterion ( $B^{k^2-2k-2}(\lambda) = \nu_k$ ), which characterises the maximiser set exactly; it is not a closed-form list of the maximisers as explicit functions of  $k$ .
- The sizes  $|\mathcal{E}_k| = 1, 6, 34, 175, 831, 3911$  ( $k = 4, \dots, 9$ ) and the lists for  $k = 4, 5$  are numerical evidence only; nothing is proved from them.
- The proof uses the Cell 1 classification of cyclic partitions as the earlier cell’s result.

## References

1. J. R. Griggs, C.-C. Ho, The cycling of partitions and compositions under repeated shifts, *Adv. Appl. Math.* **21** (1998) 205–227, DOI 10.1006/aama.1998.0597. Preprint (2 March 1998): <https://people.math.sc.edu/griggs/cycling.pdf>.

2. J. Brandt, Cycles of partitions, *Proc. Amer. Math. Soc.* **85** (1982) 483–486 (cited via Griggs–Ho Thm 2.1; not opened).